

Results: The mean residual error in the 2D inter-observer data compared with the 3D automated process was -0.11 cm (95% CI -0.41 cm/+0.19 cm), -0.21 cm (95% CI -0.39/-0.03) and -0.10 cm (95% CI -0.28/+0.08) in the left/right, anterior/posterior and superior/inferior planes. The magnitude of the residual error could be considered to be small but the 95% CI shows significant inter-observer variation that may have a clinical impact in the high dose IMRT patients due to inappropriate interventions on a daily basis.

Conclusions: The reduction of residual error is particularly significant in IMRT to enable the accurate delivery of planned dose. This study clearly identified some of the reliability issues of the 2D process for IMRT of the head and neck region. The range rather than the mean of the interpretation of the 2D data sets compared with the automated 3D process is significant. Hence, the decision was made at the Austin to implement the sole use of 3D CBCT positional verification for patients receiving IMRT for head and neck cancer and a subsequent clinical protocol was developed.

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2550 Long-term Outcomes for T4 Larynx Cancer

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Purpose/Objective(s): To evaluate long-term survival and patterns of failure for T4 larynx cancer.

Materials/Methods: Retrospective data was reviewed for sequential patients treated for T4, N0-3, M0 larynx squamous cancer between 1983 and 2004. Demographic, staging, and treatment features were recorded, and survival and disease control parameters calculated. Univariate (UV) and multivariate (MV) factors (age, gender, ethnicity, ECOG performance status, N0 vs. N+, supraglottic vs. glottic/transglottic, definitive radiation therapy (RT) vs. total laryngectomy with post-laryngectomy radiation (PLRT), biologically equivalent dose (BED), and concurrent/induction chemotherapy vs. none) were evaluated for association with overall (OS) and disease specific survival (DSS) using Cox proportional hazards modeling.

Results: There were 157 cases with median age 56 years (range, 32-79); 125 (80%) were men. 106 (66%) were N+. There were 100 (64%) supraglottic and 57 (36%) glottic/transglottic tumors. Median follow-up for all survivors was 61 months. 31 (20%) had definitive RT, 21 (67%) of whom had chemotherapy, and 126 pts (80%) had PLRT, 6 (5%) of whom had chemotherapy. Median BED delivered was 70 Gy, with a median BED of 78 Gy (SD 11), and 66 Gy (SD 6) for PLRT. 5- and 10-year OS was 57% and 37%. 5- and 10-year DSS was 63% and 54%. Local control was maintained in 128 pts (87%), locoregional control in 113 pts (77%) and freedom from recurrence in 75 pts (48%). Rates of distant metastases and 2nd primaries were 8% and 4%, respectively. In both UV and MV analysis of OS, only nodal positivity and supraglottic disease were significantly associated with mortality, and for DSS only nodal positivity (both UV $p < 0.01$, MV $p = 0.03$). UV and MV analysis of locoregional recurrence free survival was positively associated with PLRT (UV and MV $p < 0.01$). 12/31 (38%) definitive RT cases had noted locoregional relapse, compared to 21/126 (17%) for PLRT ($\chi^2 p < 0.01$). None were recorded as salvaged. Median OS for PLRT and RT cases were 87 months (CI 61-118) and 45 months (CI 27-155; $p = ns$). PLRT and RT-alone 5-year DSS were 63% and 48% ($p = ns$). RT patients had tracheotomy and reported aspiration event rates of 40% and 29%, respectively. Feeding tube rates at last contact were 19% for RT compared to 4% for PLRT ($p < 0.01$).

Conclusions: T4 larynx cancer patients can enjoy excellent cancer control and survival. The number of RT patients is too small to identify multivariate factors associated with success, or to demonstrate differences in OS/DSS between definitive and TL with post-op RT. Extreme care should be exercised in selecting patients for non-surgical treatment because of significant rates of functional impairment and locoregional failure without meaningful salvage options.

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2551 Alternative Dose for Choroidal Melanoma Treated with an Iodine 125 Radioactive Plaque: A Single Institution Retrospective Study

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Purpose/Objective(s): The Collaborative Ocular Melanoma Study (COMS) established radioactive iodine eye plaques as an accepted standard of care in the treatment of medium sized choroidal melanoma. In the COMS study, "standard" prescribed doses were 85 Gy. This retrospective review was performed to study the outcomes of patients treated with lower than "standard" doses.

Materials/Methods: From 1990 to 2004, 63 patients underwent I-125 episcleral plaque application (per COMS design) for treatment of choroidal melanoma. For purposes of statistical analysis, patient, tumor and treatment variables were evaluated in terms of patient outcomes including enucleation rates, visual results, local control, disease free survival, cause-specific survival, overall survival, and treatment related effects. Time to event (survival, visual acuity (VA) loss) outcomes was estimated using the Kaplan-Meier method. Cox Proportional Hazards regression was performed using backwards selection of correlates significant in univariate analysis.

Results: Median follow-up time on the 63 patients was 4.8 yrs (58.2 mos.). The mean tumor height was 4.6 ± 1.9 mm. The average plaque size was 15.1 ± 2.4 mm and the average dose rate at tumor apex was 62.7 ± 7.4 cGy/hour. The average total dose was 62.5 ± 2.8 Gy (range 56 - 69 Gy) and the average treatment depth was 6.0 ± 1.3 mm. There were 7 local failures and 4 distant failures. During the follow-up period, there were ten enucleations (6 secondary to local failure and 4 secondary to treatment complications). At 5 years, overall survival was 76.8%, disease-free survival was 94.4%, cause-specific survival was 95.9%, local control was 88.1%, enucleation rate was 13.8% and visual acuity (VA) better than 20/200 was 76.4%, respectively. The only significant predictors of local failure were blurred vision and visual field defect (VFD) at presentation ($p = 0.0052$). Diabetic retinopathy was associated with poorer outcome in time to VA $<20/200$ (HR = 39.1, $p < 0.0001$), poorer initial VA in the tumor eye (HR = 0.09, $p = 0.0037$) and relative afferent papillary defect (RAPD) (HR = 60.2, $p = 0.0004$).